

k-NN, SVM, and Neural Nets in Practice

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*“... what we want is a machine that learns from experience.”
~ Alan Turing*

Abstract—This paper evaluates the performance of three supervised machine learning algorithms, k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Neural Networks (NN), on two structurally different datasets: Global Cancer Patients and Company Bankruptcy. The goal is to understand how algorithm performance varies across tasks with fundamentally different characteristics. For the regression task on the cancer dataset, the objective was to predict the severity score of a patient. For the classification task on the bankruptcy dataset, the aim was to predict whether a company will go bankrupt.

We hypothesized that NN would outperform both k-NN and SVM due to their capacity to model complex patterns and generalize well across diverse problem types. Our experiments confirm this hypothesis: NN consistently achieved superior performance in both regression and classification settings. These results suggest that neural networks are highly adaptable and effective across varied domains, provided appropriate tuning and sufficient data.

I. INTRODUCTION

Machine learning, a subfield of artificial intelligence, enables computers to learn patterns from data without being explicitly programmed for each task. Rather than mimicking human consciousness, machine learning algorithms infer patterns and relationships from datasets to make predictions or decisions [1]. A core branch of machine learning is supervised learning, where models are trained on labeled data: each input feature vector is paired with a corresponding output value.

This paper focuses on evaluating three widely-used supervised learning algorithms, k-Nearest Neighbors (kNN), Support Vector Machines (SVM), and Artificial Neural Networks (ANN), on two fundamentally different datasets. These algorithms are capable of handling both regression problems (predicting continuous values) and classification problems (predicting discrete labels). kNN predicts by aggregating the outputs of the closest data points, using mean for regression and majority voting for classification. SVM attempts to find an optimal hyperplane that separates different classes with the maximum margin, and in regression settings, it minimizes deviations within a set margin. Neural networks are composed of interconnected neurons inspired by biological neural systems, and are well-suited for learning complex nonlinear patterns.

The first dataset, the Global Cancer Patient dataset from Kaggle, presents a unique challenge: it contains a mix of categorical and numerical features with varying scales in a high-dimensional space. This complexity makes the dataset interesting from a machine learning perspective, as it requires careful preprocessing, such as one-hot encoding of

categorical features and standardization of numerical ones, to enable models to learn effectively. Furthermore, the target variable is a multi-class score representing patient severity, making this a regression problem with nuanced patterns to uncover.

The second dataset, Company Bankruptcy, consists entirely of numerical features but also exhibits high dimensionality and imbalanced class distribution. This makes it an ideal candidate to test model robustness in a binary classification context, especially in the presence of skewed labels.

The **hypothesis** is that Artificial Neural Networks will outperform both kNN and SVM across both datasets. For the cancer dataset, we expect ANN to excel due to its ability to model nonlinear relationships and handle heterogeneous data types. SVM with an RBF kernel is expected to follow, as it can capture nonlinear decision boundaries. We hypothesize that kNN will perform the worst, due to its poor generalization in high-dimensional spaces and sensitivity to irrelevant or categorical features.

II. METHODOLOGY

This experiment was conducted to evaluate how different machine learning algorithms perform across datasets with varying characteristics. The datasets used were obtained from Kaggle, a well-known platform for data science competitions and community-contributed datasets.

The first dataset, Global Cancer Patients, contains information on cancer patient cases collected between 2015 and 2024. This dataset includes both numerical and categorical features, requiring multiple preprocessing steps. In particular, categorical columns were transformed using one-hot encoding, a common technique to convert non-numeric values into numerical format suitable for model training. This transformation is essential because most machine learning algorithms operate on numerical input, whether calculating distances in kNN or constructing hyperplanes in SVM. Additionally, standardization (feature scaling) was applied to normalize the numerical values, ensuring that features with larger ranges did not disproportionately influence model training.

The second dataset, Company Bankruptcy, contains financial and operational information about companies and indicates whether a company went bankrupt. This dataset is fully numerical, so the only preprocessing step needed was standardization. No other transformations or feature engineering techniques were applied.

The Global Cancer Patients dataset initially contained 50,000 rows and 15 columns. One of the non-numeric columns, ‘Patient ID’, was dropped because it was a unique identifier and did not contribute useful information for prediction. Of the remaining features, four were categorical: ‘Gender’, ‘Country Region’, ‘Cancer Type’, and ‘Cancer Stage’. These were transformed using one-hot encoding, resulting in a total of 34 features used for model training after preprocessing.

The Company Bankruptcy dataset contained 6,819 rows and 96 numerical features. All columns were retained, and no features were removed. The only transformation applied was standardization.

The target variable in the Global Cancer Patients dataset was the Target Severity Score, a continuous value, framing the task as a regression problem. In contrast, the Company Bankruptcy dataset used the binary variable Bankrupt? as the target, making it a classification problem.

III. RESULTS

A. *k*-Nearest Neighbors (*k*-NN)

k-Nearest Neighbors (*k*-NN) is a non-parametric, instance-based learning algorithm that does not construct a model during training. Instead, it memorizes the training data and makes predictions by comparing new inputs to their nearest neighbors using a distance metric. For regression, *k*-NN predicts the average output of the *k* nearest neighbors [2]. For classification, it uses a majority vote. This experiment used Euclidean distance to compute similarity.

The main hyperparameter tuned for *k*-NN is *k*, the number of neighbors. A small *k* can lead to high variance and overfitting, while a large *k* can increase bias by oversmoothing decision boundaries. This trade-off is explored in both regression and classification contexts.

a) Global Cancer Patients (Regression): For the cancer dataset, the loss function used was Root Mean Squared Error (RMSE), selected for its sensitivity to large errors, appropriate when severe mispredictions could be costly. Figure 1 shows the learning curve with *k* = 5. The training RMSE consistently remains lower than the validation RMSE across increasing training sizes, indicating overfitting. The gap between the two curves does not narrow significantly, suggesting that adding more data alone may not be sufficient to close the generalization gap.

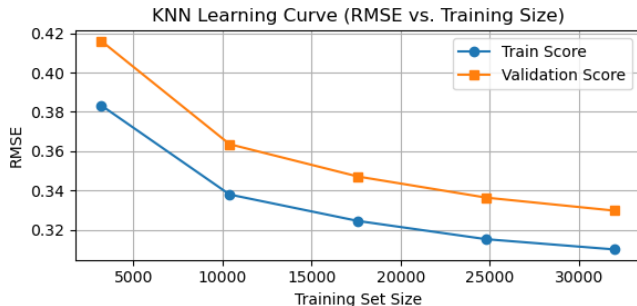


Fig. 1. *k*-NN Learning Curve (RMSE vs. Training Size) for *k* = 5

This overfitting is further supported by the validation curve in Figure 2, which plots RMSE against various *k* values. At *k* = 1, the model memorizes the training set, achieving near-zero RMSE on training but high RMSE on validation. As *k* increases, training error rises, while validation error decreases initially, demonstrating a textbook bias-variance trade-off. The curve flattens around *k* = 10, indicating diminishing returns in generalization with further increases in *k*.

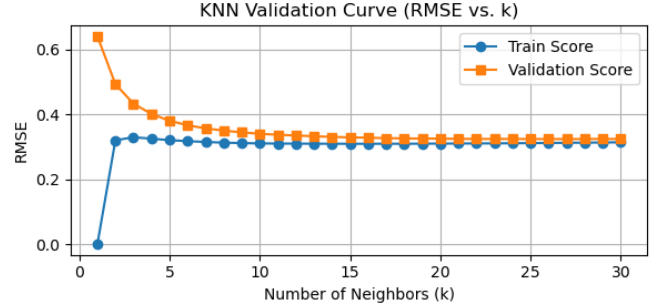


Fig. 2. *k*-NN Validation Curve (RMSE vs. *k*) for Global Cancer Dataset

b) Company Bankruptcy (Classification): In the classification setting, the loss metric used was the F1 score, an appropriate choice given the severe class imbalance (bankrupt vs. non-bankrupt). Figure 3 shows the learning curve. As the training set grows, both training and validation F1 scores improve, but the improvement plateaus around an F1 of 0.29 on validation data, suggesting that the model struggles to capture signals from the minority class.

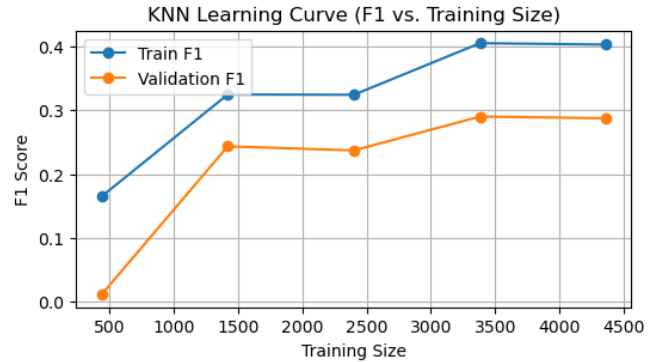


Fig. 3. *k*-NN Learning Curve (F1 vs. Training Size) for Bankruptcy Dataset

This weakness is also evident in the classification report and confusion matrix. The model achieves an overall accuracy of 97%, but this is misleading: the F1 score for the minority class is only 0.31 due to poor recall (0.20), showing that the model rarely identifies bankrupt companies correctly.

Figure 4 shows the validation curve. The optimal *k* lies around 3 to 5, aligning with earlier observations. At *k* = 1, the model overfits, achieving a perfect training F1 but a very low validation F1. Beyond *k* = 10, both training and validation performance degrade, indicating the model becomes too simplistic to capture rare class patterns.

Class	Precision	Recall	F1-Score	Support
0 (Not Bankrupt)	0.97	1.00	0.99	1320
1 (Bankrupt)	0.64	0.20	0.31	44
Accuracy	0.97			
Macro Avg	0.81	0.60	0.65	1364
Weighted Avg	0.96	0.97	0.96	1364

TABLE I
CLASSIFICATION REPORT FOR K-NN ON BANKRUPTCY DATASET

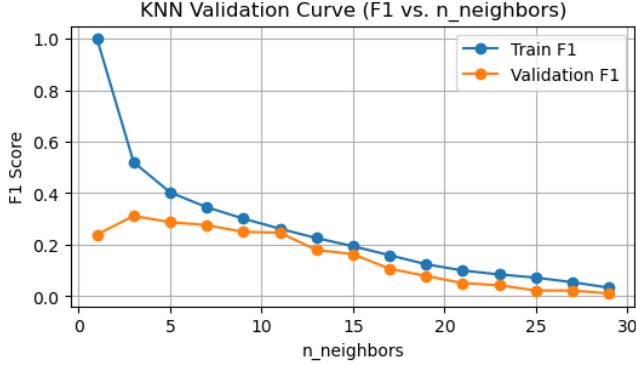


Fig. 4. k-NN Validation Curve (F1 vs. k) for Bankruptcy Dataset

c) Efficiency Considerations: Training time for k-NN was negligible due to its lazy-learning nature. For the cancer dataset regression task, training took only 0.0021 seconds, while prediction took 0.41 seconds, highlighting k-NN's key trade-off: fast training but slow inference. In classification, the bankruptcy model trained in 0.0072 seconds, with prediction time of 0.5719 seconds.

d) Summary: Overall, k-NN underperformed on both datasets. It overfit the cancer dataset and failed to generalize well in the bankruptcy dataset, especially in detecting rare positive cases. These results support the initial hypothesis: k-NN is not well-suited to high-dimensional, mixed-type, or imbalanced datasets, and is outperformed by more sophisticated models like SVM and neural networks.

B. Support Vector Machines (SVM)

Support Vector Machines (SVMs) are supervised learning algorithms that find an optimal hyperplane to separate data in classification tasks or predict continuous values in regression. A key strength of SVMs lies in their ability to use kernel functions to project data into higher-dimensional spaces where linear separation becomes feasible [3].

This study evaluated both Radial Basis Function (RBF) and Linear kernels across two tasks: support vector regression (SVR) for the Global Cancer Patient dataset, and support vector classification (SVC) for the Company Bankruptcy dataset.

a) Global Cancer Patients (Regression): Figure 5 compares SVR learning curves for both kernels. The RBF kernel shows strong improvements in both training and validation RMSE as the training set increases. Training RMSE decreases from 0.073 to 0.049, while validation RMSE drops

from 0.122 to 0.057, highlighting reduced overfitting and better generalization with more data. The narrowing gap indicates the model benefits from the increased data capacity to learn nonlinear patterns.

In contrast, the linear kernel achieves lower error overall, validation RMSE drops to 0.0335, with near-identical training and validation scores. This suggests that the relationship between features and severity score is sufficiently linear after preprocessing, and additional model complexity from RBF is unnecessary.

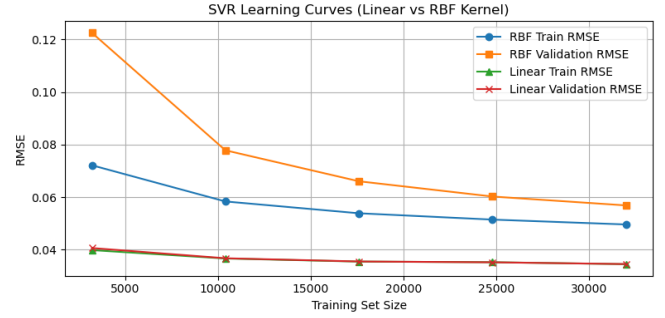


Fig. 5. SVR Learning Curves for Global Cancer Dataset (Linear vs RBF Kernel)

Validation curves (Figure 6) reinforce this observation. For the RBF kernel, model performance improves with higher regularization strength C , but plateaus around $C = 1$, suggesting a sweet spot in the bias-variance tradeoff. The linear kernel performs best with $C = 0.1$, beyond which no significant improvement is seen and potential overfitting may occur.

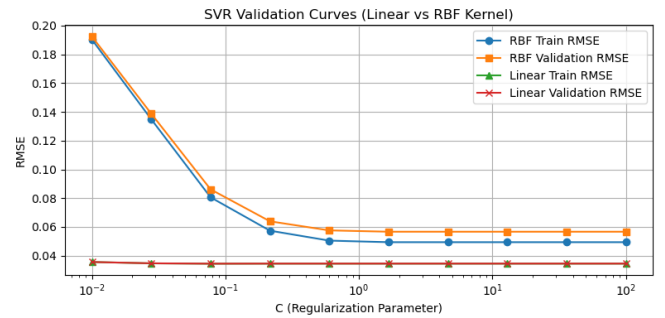


Fig. 6. SVR Validation Curves (RMSE vs. C) for Global Cancer Dataset

Among SVR models, the linear kernel achieved the best results:

- **SVR (Linear):** RMSE = 0.0335, MAE = 0.0271, R^2 = 0.9992, Train Time = 0.61s
- **SVR (RBF):** RMSE = 0.0537, MAE = 0.0404, R^2 = 0.9980, Train Time = 21.87s

This indicates that the linear SVR not only performs better but is also significantly faster to train, validating the hypothesis that simpler models can outperform more complex ones when the data supports linearity.

b) *Company Bankruptcy (Classification)*: Figure 7 shows the F1-score learning curves for both kernels. The RBF kernel outperforms the linear kernel on validation F1 (peaking at 0.31 vs. 0.28), consistent with the hypothesis that RBF's ability to capture nonlinear boundaries is valuable in imbalanced, high-dimensional data. However, neither kernel achieves high generalization performance, as both show significant gaps between training and validation scores.

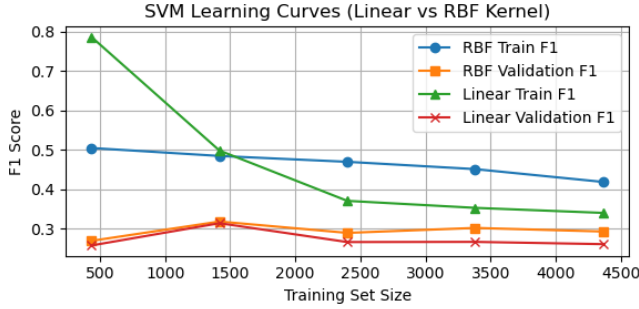


Fig. 7. SVC Learning Curves (F1 vs. Training Size) for Bankruptcy Dataset

Validation curves in Figure 8 demonstrate the effect of regularization. For both kernels, increasing C boosts training F1 scores, but has limited impact on validation performance, suggesting overfitting at high values of C . RBF shows the strongest F1 improvement around $C = 1$, while the linear kernel peaks near $C = 5$.

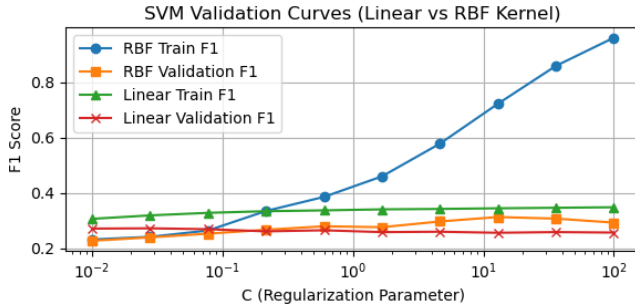


Fig. 8. SVC Validation Curves (F1 vs. C) for Bankruptcy Dataset

Final classification reports highlight this difference:

SVM (RBF Kernel):

- F1 Score (Class 1 - Bankrupt): 0.31
- Recall (Class 1): 0.55
- Accuracy: 92.1%
- Prediction Time: 0.53s

SVM (Linear Kernel):

- F1 Score (Class 1 - Bankrupt): 0.28
- Recall (Class 1): 0.80 (highest recall, but very low precision)
- Accuracy: 87.0%
- Prediction Time: 0.12s

The linear kernel achieved better recall but suffered from very low precision, misclassifying many non-bankrupt companies. This indicates overcompensation for imbalance,

where the model sacrifices precision for recall. RBF balanced both, making it better overall for minority class detection, aligning with our hypothesis that RBF would outperform linear and k-NN under imbalanced settings.

c) *Training Efficiency*: SVR with the RBF kernel had significantly longer training times due to its nonlinear mapping and high-dimensional optimization complexity. In contrast, the linear kernel was faster and more efficient in both regression and classification tasks. SVM classification generally trained and predicted faster due to the smaller dataset size and binary target variable.

d) *Summary*: SVM models, especially those with RBF kernels, performed well on both datasets, with strong generalization and better handling of imbalanced classification compared to k-NN. On the cancer dataset, the linear SVR surprisingly outperformed RBF SVR, likely due to linear relationships after preprocessing. These results partially confirm our hypothesis: SVM with nonlinear kernels is powerful in complex feature spaces, but linear models can still excel when the data permits.

C. Neural Networks (MLP)

Artificial Neural Networks (ANNs), inspired by the structure of the human brain, consist of interconnected units (neurons) organized in layers [4]. These models learn non-linear mappings from input features to outputs by optimizing weights through backpropagation. For this experiment, a Multi-Layer Perceptron (MLP) was used with two activation functions: ReLU and tanh, across both regression and classification tasks [4].

a) *Global Cancer Dataset (Regression)*: Figure 9 shows learning curves comparing ReLU and Tanh activations. Both configurations show significant improvement in RMSE with increasing training size. The training and validation curves converge tightly, indicating low variance and strong generalization. This confirms the hypothesis that neural networks excel on structured, high-dimensional data.

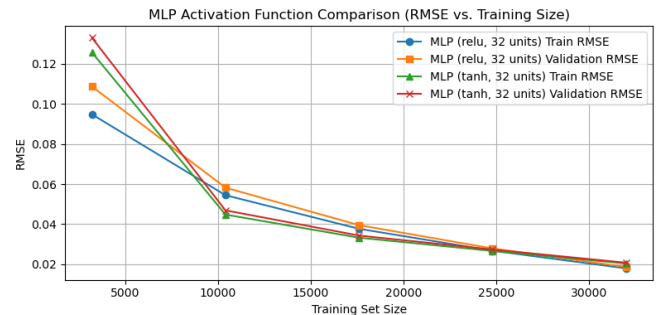


Fig. 9. MLP Learning Curves (RMSE vs. Training Size), ReLU vs. Tanh

Both models achieve test RMSE of 0.012 with $R^2 = 0.9999$, reflecting near-perfect predictive performance:

- ReLU: RMSE = 0.0156, MAE = 0.0117, $R^2 = 0.9998$
- Tanh: RMSE = 0.0126, MAE = 0.0082, $R^2 = 0.9999$

b) Company Bankruptcy Dataset (Classification): For classification, MLP exhibited classic overfitting behavior. Training F1 scores were consistently near 0.99 for both activations, but validation F1 peaked around 0.35 before plateauing, as seen in Figure 10. This suggests that model complexity enables excellent training performance, but dataset imbalance and input feature limitations hinder generalization.

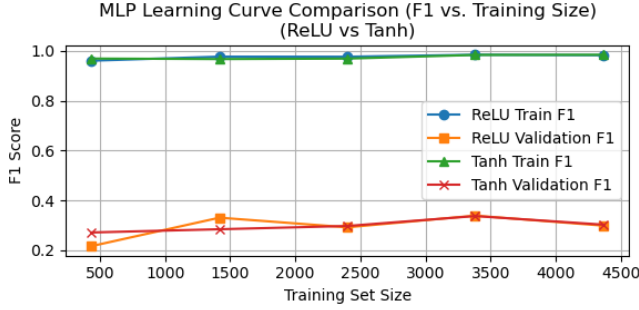


Fig. 10. MLP Learning Curves (F1 vs. Training Size) , ReLU vs. Tanh

c) Model Complexity and Hidden Layer Size: Validation curves (Figure 11) demonstrate that performance is optimal when the hidden layer contains 16–32 neurons. Fewer neurons result in underfitting, while overly large networks (64+) lead to overfitting. This pattern holds for both activation functions, illustrating the bias-variance trade-off.

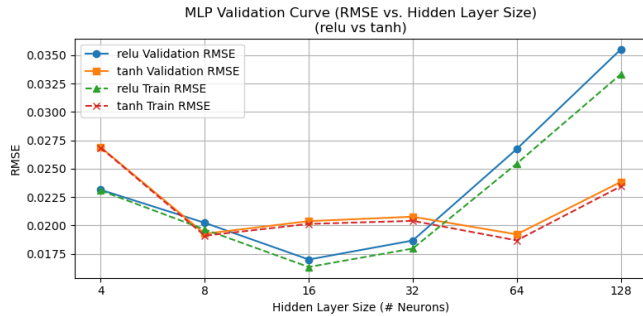


Fig. 11. Validation Curve (RMSE vs. Hidden Size) , Global Cancer Dataset

A similar effect is observed in the classification task, where validation F1 peaks at 32 neurons (Figure 12). Beyond this point, performance declines due to overfitting, confirming the need to control model capacity under class imbalance.

d) Training Convergence: Figures 13 and 14 show iterative learning curves. For regression, RMSE flattens after 50 epochs, indicating early convergence. For classification, training F1 improves rapidly, while validation F1 stabilizes around 0.35–0.40. Tanh shows slightly higher variance in validation performance, while ReLU produces smoother generalization.

e) Weight Initialization: In regression, uniform initialization outperformed Xavier and He (Figure 15). This is expected, as Xavier and He are more beneficial in deeper networks, while the architecture here is relatively shallow. For classification (Figure 16), both Xavier and Uniform

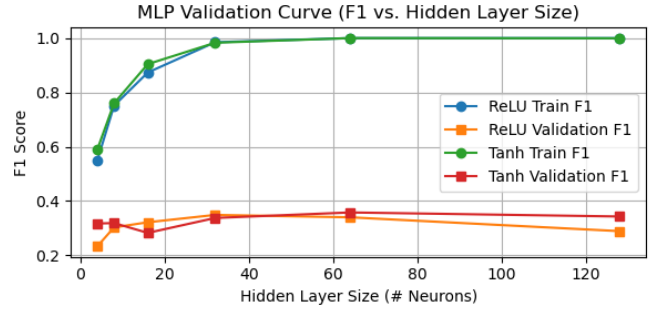


Fig. 12. Validation Curve (F1 vs. Hidden Size) , Bankruptcy Dataset

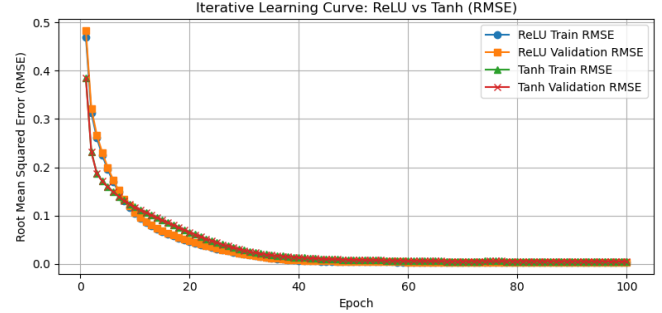


Fig. 13. MLP Iterative Learning Curve (RMSE vs. Epochs) , Global Cancer Dataset

performed similarly well, while He lagged, possibly due to sensitivity to ReLU's behavior on small data.

f) Architecture Comparison: Figure 17 shows that deep narrow and baseline architectures outperform shallow wide ones in regression, with RMSE around 0.011. This emphasizes the value of depth over width for capturing complex relationships. In classification (Figure 18), all architectures perform similarly, with marginal F1 improvements, highlighting the difficulty imposed by the dataset's imbalance.

g) L2 Regularization: Adding L2 regularization slightly increased RMSE in regression (Figure 19), suggesting no benefit under this already-regularized setup. In classification (Figure 20), however, L2 marginally improved the F1 score (0.0642 vs. 0.062), hinting at better stability.

h) Final Evaluation: Global Cancer Dataset (Regression):

- Final Model (Tanh): RMSE = 0.0126, $R^2 = 0.9999$, Train Time = 2.35s, Predict Time = 0.0017s
- Final Model (ReLU): RMSE = 0.0156, $R^2 = 0.9998$, Train Time = 3.60s, Predict Time = 0.0122s

Company Bankruptcy (Classification):

- ReLU: F1 (minority) = 0.30, Accuracy = 95.97%, Train Time = 4.57s, Predict = 0.0007s
- Tanh: F1 (minority) = 0.41, Accuracy = 96.41%, Train Time = 5.54s, Predict = 0.0018s

i) Summary: Neural networks outperformed both SVM and k-NN in regression tasks, achieving near-perfect scores. In classification, while performance was still limited by data

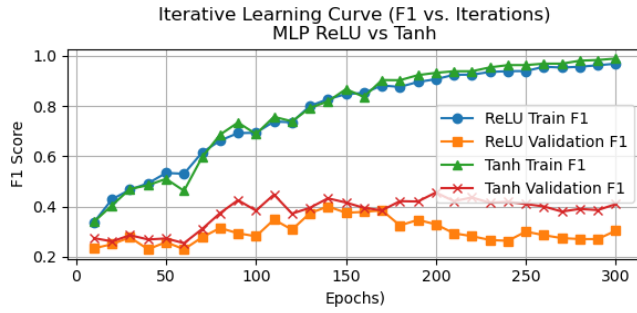


Fig. 14. MLP Iterative Learning Curve (F1 vs. Epochs), Bankruptcy Dataset

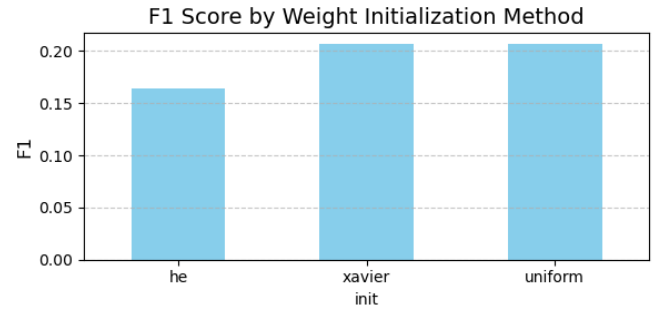


Fig. 16. Weight Initialization Comparison (F1), Bankruptcy Dataset

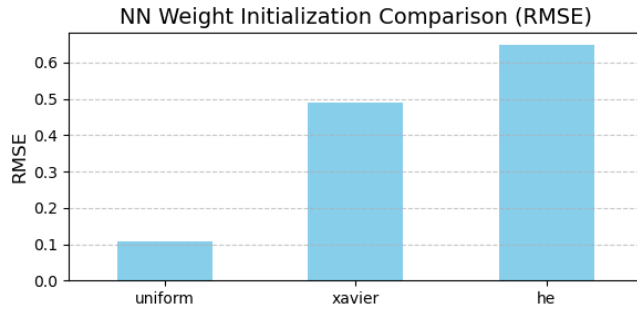


Fig. 15. Weight Initialization Comparison (RMSE), Global Cancer Dataset

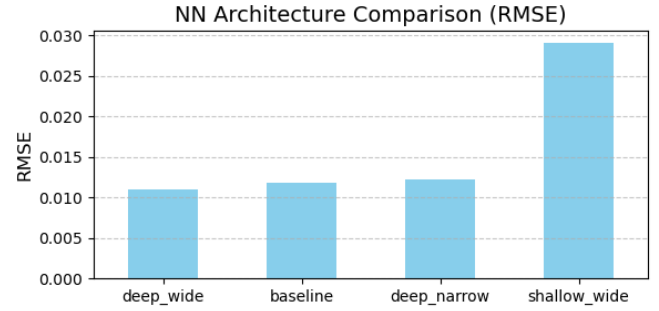


Fig. 17. MLP Architecture Comparison (RMSE), Global Cancer Dataset

imbalance, MLP surpassed k-NN and matched or slightly exceeded SVM, validating the original hypothesis. The experiments further revealed that careful tuning of architecture, activation, and initialization significantly affects generalization, particularly in imbalanced classification settings.

IV. CONCLUSION

In this project, we conducted a comparative analysis of three supervised machine learning algorithms, k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Neural Networks (NN), on two contrasting datasets. The Global Cancer Patient dataset framed a regression task, while the Company Bankruptcy dataset represented a binary classification problem. Beyond evaluating predictive performance, we examined how model behavior is influenced by hyperparameter tuning, class imbalance, data characteristics, and computational efficiency.

Our results demonstrate that SVM and NN excel at capturing complex, nonlinear relationships when appropriately configured. The Multi-Layer Perceptron (MLP) achieved near-perfect predictive accuracy on the regression task ($R^2 > 0.99$), validating our hypothesis that neural networks perform well on high-dimensional, mixed-type data. However, on the classification task, MLP's performance was hindered by severe class imbalance and overfitting, despite extensive tuning.

SVM also showed strong performance, especially with the RBF kernel in classification, confirming its effectiveness in learning nonlinear decision boundaries. Still, its performance depended heavily on the choice of kernel and regularization

parameters.

In contrast, k-NN offered simplicity and interpretability but struggled with scalability, high-dimensional data, and imbalanced classes. While k-NN occasionally performed competitively, especially in low-bias settings, it was consistently outperformed by the other models, supporting our initial hypothesis.

This project highlights the importance of aligning model complexity with data structure, rigorously tuning hyperparameters, and accounting for real-world dataset challenges such as imbalance. Ultimately, while neural networks showed the highest ceiling of performance, especially in regression, no single model dominated in all cases, underscoring the need for thoughtful algorithm selection tailored to problem characteristics.

References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

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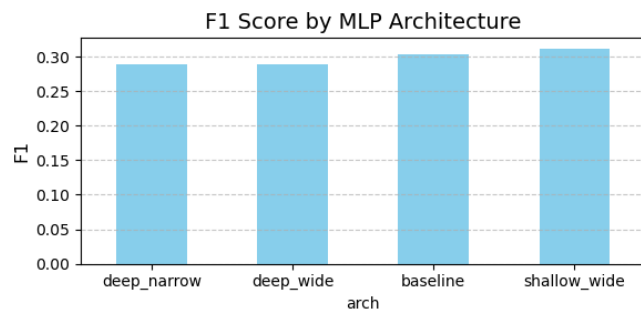


Fig. 18. MLP Architecture Comparison (F1) , Bankruptcy Dataset

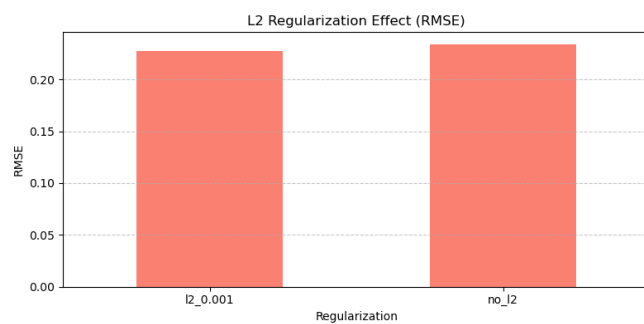


Fig. 19. L2 Regularization Effect (RMSE) , Global Cancer Dataset

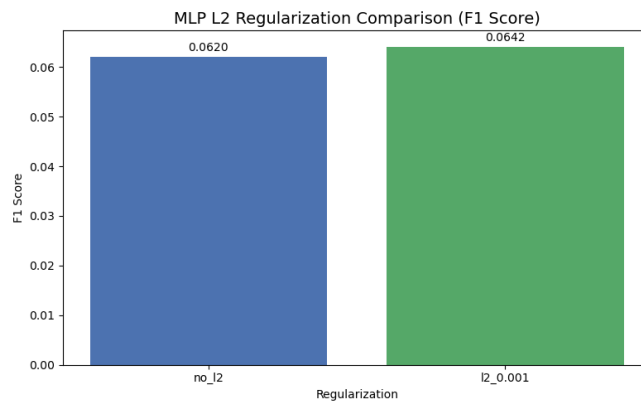


Fig. 20. L2 Regularization Effect (F1) , Bankruptcy Dataset